

Pydantic Interview Questions — Basic to Intermediate

Basic

1. What is Pydantic?
2. Why do we need Pydantic in FastAPI?
3. What is BaseModel in Pydantic?
4. How do you create a Pydantic model?
5. How does Pydantic perform data validation?
6. What happens when invalid data is passed to a Pydantic model?
7. What is the difference between **Pydantic model** and a **normal Python class**?
8. What are required and optional fields in Pydantic?
9. How do you define a field with a default value?
10. How is Pydantic used with FastAPI?
11. How does FastAPI automatically validate request-body data using Pydantic?
12. What is a request body in FastAPI?
13. What is path parameter, query parameter, and request body?
14. What is the difference between Pydantic validation and manual if-else validation?

Intermediate

15. How do you make a field optional?
16. What is the difference between `Optional[str]` and `str | None`?
17. What is `Field()` in Pydantic?
18. How do you add constraints such as minimum/maximum length or value?
19. How do you validate an email address in Pydantic?
20. What are Pydantic validators?
21. What is the difference between field validation and model validation?
22. How do you handle validation errors in FastAPI?
23. How do you convert a Pydantic model into a dictionary?

- 24. How do you create a Pydantic model from dictionary/JSON data?
- 25. How do you use Pydantic models for **POST, PUT, and PATCH** requests?
- 26. How do Pydantic models work with MongoDB in a FastAPI application?
- 27. What is `model_dump()` in Pydantic?

Practical / Logical Interview Questions

- 28. If a user sends age: "25" when age is defined as int, what happens?
- 29. What happens if a required field is missing from the request?
- 30. How would you validate that a user's age must be between 18 and 60?
- 31. How would you validate that an email has a valid format?
- 32. How would you create a Pydantic model for a nested address object?
- 33. How would you create a custom validation rule, such as password and `confirm_password` needing to match?
- 34. If invalid data is sent through Postman, how does FastAPI + Pydantic handle it?